

LECTURE 5

Topics

Diversifiable or unsystematic risk, Undiversifiable or systematic risk, Achieving diversification through portfolios, Index models, Treynor-Black model of optimal portfolio allocation when assets have a single index structure

- Consider a simplified version of the portfolio analysis performed in Lecture 4:
 - There are n risky assets and each asset, P_i , has the same return volatility:

$$\sigma_{P_i} = \bar{\sigma}$$

- The covariance between any two risky assets, P_i and P_j , is the same:

$$\sigma_{P_i P_j} = \underline{\sigma}^2$$

- The variance-covariance matrix of these assets becomes:

$$\Sigma = \begin{pmatrix} \bar{\sigma}^2 & \underline{\sigma}^2 & \dots & \underline{\sigma}^2 \\ \underline{\sigma}^2 & \bar{\sigma}^2 & \dots & \underline{\sigma}^2 \\ \dots & \dots & \dots & \dots \\ \underline{\sigma}^2 & \underline{\sigma}^2 & \dots & \bar{\sigma}^2 \end{pmatrix}$$

- Consider an equally-weighted portfolio, S , of n risky assets:

$$w_S = \left(\frac{1}{n} \quad \frac{1}{n} \quad \dots \quad \frac{1}{n} \right)$$

- Then the variance of this portfolio can be calculated with the formula:

$$\sigma_S^2 = w_S \times \Sigma \times w_S^T$$

which simplifies to:

$$\sigma_S^2 = \frac{\bar{\sigma}^2}{n} + \frac{n-1}{n} \underline{\sigma}^2$$

- Note that most of σ_S comes from $\underline{\sigma}^2$ and a little from $\bar{\sigma}^2$
- When n is sufficiently large:
 - $\frac{1}{n} \rightarrow 0$, and $\bar{\sigma}$ has negligible effect on σ_S
 - $\frac{n-1}{n} \rightarrow 1$, and $\underline{\sigma}$ becomes the main effect in σ_S

- The main takeaway is that with more assets, the risk of the portfolio actually goes *down*
- This is known as the *portfolio diversification effect*:

Portfolio diversification effect

As the number of assets in a portfolio increases, the contribution of an individual asset to the risk of the portfolio reduces to its covariance with the other assets in the portfolio. The "idiosyncratic" portion of the individual asset's risk is *diversified* away.

- The portion of an individual asset's risk that captures the covariance with other assets is known as *undiversifiable* or *systematic* risk
- The remaining portion of the individual asset's risk is "idiosyncratic" (asset specific) and is known as *diversifiable* or *unsystematic* or simply *idiosyncratic* risk

- The portfolio diversification analysis above suggests that we can view the total risk of an individual asset as having two components:
 - Idiosyncratic or diversifiable or nonsystematic risk, that captures firm specific
 - Non-diversifiable or systematic risk that captures exposures to common sources of risk that affect all assets in the economy
- How do we identify the diversifiable and non-diversifiable components of an individual assets' risk?
- If we knew the tangency portfolio (optimal risky portfolio), T :
 - The non-diversifiable risk of asset i is $\frac{\text{cov}^2(r_i, r_T)}{\sigma_T^2}$
 - The diversifiable risk of asset i is the residual: $\sigma_i^2 - \frac{\text{cov}^2(r_i, r_T)}{\sigma_T^2}$

- We noted in Lecture 4 that finding the tangency portfolio T in practice gets quickly complicated because of the large number of parameters that have to be estimated (e.g., most of them come from the variance-covariance matrix)
- We can take a simpler approach that recognizes that the reason why risky assets comove together (i.e., covariance) is because they respond to unexpected changes in macroeconomic variables
- For example: business cycles (i.e., aggregate demand), interest rate risk, cost of common inputs of production
- We can attempt to measure how a risky asset responds to changes in each macro variable
- However, many macro variable are only available at low frequencies, such as quarterly, semiannually or annually
- At the same time, the returns on risky assets can be calculated at high frequencies, such as monthly, weekly or daily

- This frequency mismatch raises the question: Can we find ways to capture changes in macro variables at higher frequencies using financial instruments?
- The answer is yes: A market index, M , will have exposure to all these macro variables
- Then we can measure how risky asset responds to changes in macro variables by measuring how it responds to changes in M
- One way to do this measurement is to explicitly link the return of a risky asset i in excess of the risk-free rate, $r_i - r_F$ to that of M :

$$r_i(t) - r_F(t) = \alpha_i + \beta_i \times (r_M(t) - r_F(t)) + e_i(t)$$

- α_i and β_i are constants
- $e_i(t)$ capture idiosyncratic (firm specific) shocks that have nothing to do with the market and that have a mean of zero, $E[e_i] = 0$
- This equation is called a *single index model*

- The coefficients α_i and β_i can be estimated from a simple regression analysis
- We use the index model to learn about the risk premium of the risky asset
- If we take the expectation, $E(\cdot)$, of both sides of the equation we obtain:

$$E(r_i - r_F) = \alpha_i + \beta_i \times (E(r_M - r_F))$$

- This equation says that the risk premium of asset i , $E(r_i - r_F)$, has two components:
 - A market component that is proportional to the risk premium of the market, $\beta_i \times (E(r_M - r_F))$
 - A non-market component, α_i , that arises when M doesn't capture well the changes in some of the macro variables

- We can also use the index model to learn about the asset's diversifiable and non-diversifiable risk components
- If we take the variance of both sides of the index model equation we obtain:

$$\sigma_i^2 = \beta_i^2 \times \sigma_M^2 + \sigma^2(e_i)$$

- This equation says that the total risk of asset i , σ_i^2 , has two components:
 - A non-diversifiable component $\beta_i^2 \times \sigma_M^2$
 - A diversifiable component: $\sigma^2(e_i) = \sigma_i^2 - \beta_i^2 \times \sigma_M^2$
- We will refer to $\sigma(e_i)$ as the *residual standard deviation*
- We now illustrate how to estimate the the index model (pls download index models data.xls from Week 10 page):
 - Market index: SPY - an ETF that tracks S&P 500
 - Stocks: IBM and Apple

- In the example above, we estimate the parameters of the index model using statical methods (e.g., regression analysis)
- However, in the finance industry, the common practice is to actually estimate each parameter *separately*
- The typical steps are as follows:
 - 1 Macroeconomic analysis: Estimate the risk premium of the market $E(r_M) - E(r_F)$
 - 2 Statistical analysis: Estimate the market beta, β_i , and the residual variances $\sigma^2(e_i)$
 - 3 Calculate the expected return of the asset coming solely from exposure to the market: $\beta_i \times (E(r_M) - E(r_F))$
 - 4 Security analysis: Forecast the asset's alpha, α_i , using proprietary valuation models and high quality data (i.e., *private information*)
 - 5 Finally, calculate the expected return of the asset:
$$E(r_i) = E(r_F) + \alpha_i + \beta_i \times (E(r_M) - E(r_F))$$

- We can use the index equation to learn about the comovement between two risky assets
- The index model applied to two risky asset i and j becomes:

$$r_i(t) - r_F(t) = \alpha_i + \beta_i \times (r_M(t) - r_F(t)) + e_i(t)$$

$$r_j(t) - r_F(t) = \alpha_j + \beta_j \times (r_M(t) - r_F(t)) + e_j(t)$$

- Recall that $e_i(t)$ and $e_j(t)$ are firm-specific shocks and should not be related to each other or to the risk-free asset or the market
- At the same time, the risk-free asset has virtually no risk:

$$\text{Cov}(e_i, e_j) = 0$$

$$\text{Cov}(e_i, r_M - r_F) = \text{Cov}(e_j, r_M - r_F) = 0$$

$$\text{Cov}(r_F, r_M) = \text{Cov}(r_F, r_F) = 0$$

- Then the covariance between assets i and j becomes:

$$\text{Cov}(r_i, r_j) = \beta_i \beta_j \sigma_M^2$$

- We can use the single index model to simplify the variance-covariance matrix of risky assets and to solve the optimal portfolio allocation problem
- The single-index model allows us to calculate expected returns and covariances using α and β parameters only:
 - Expected return:

$$E(r_i) = E(r_F) + \alpha_i + \beta_i \times (E(r_M) - E(r_F))$$

- Variance of an asset i :

$$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma^2(e_i)$$

- Covariance between two assets i and j :

$$\text{Cov}(r_i, r_j) = \beta_i \beta_j \sigma_M^2$$

- Covariance between asset i and the market:

$$\text{Cov}(r_i, r_M) = \beta_i \sigma_M^2 = \beta_i \sigma_M^2$$

- With these expected returns and covariances, we can apply the portfolio allocation analysis in Lecture 4
- We start with the optimal portfolio allocation for n assets
- Because the index model relies on the market as being one of the traded assets, we include the market M in our set of assets
- We start by constructing the expected returns and variance-covariance matrices
- Expected returns:

$$E(r) = (E(r_1) \quad E(r_2) \quad \dots \quad E(r_n) \quad E(r_M))$$

- Return variance covariance matrix:

$$\Sigma = \begin{pmatrix} \beta_1^2 \sigma_M^2 + \sigma^2(e_1) & \beta_1 \beta_2 \sigma_M^2 & \dots & \beta_1 \beta_n \sigma_M^2 & \beta_1 \sigma_M^2 \\ \beta_2 \beta_1 \sigma_M^2 & \beta_2^2 \sigma_M^2 + \sigma^2(e_2) & \dots & \beta_2 \beta_n \sigma_M^2 & \beta_2 \sigma_M^2 \\ \dots & \dots & \dots & \dots & \dots \\ \beta_1 \sigma_M^2 & \beta_1 \sigma_M^2 & \dots & \beta_n \sigma_M^2 & \sigma_M^2 \end{pmatrix}$$

- We now consider a portfolio S that invests in the n assets and the market:

$$w_S = (w_1 \quad w_2 \quad \dots \quad w_n \quad w_M)$$

- The expected return of portfolio S becomes:

$$E(r_S) = E(r) \times w_S^T$$

- The return volatility of portfolio S becomes:

$$\sigma_S = \sqrt{w_S \times \Sigma \times w_S^T}$$

- We use these two to calculate the Sharpe's ratio:

$$SR_S = \frac{E(r_S) - E(r_F)}{\sigma_S}$$

- Finally, we use Excel to find the portfolio S with the largest SR_S
- This is the optimal risky portfolio, T

- When using index models, the optimal risky portfolio T can also be found via calculus
- The advantage to this approach is that we can actually calculate T with a formula
- Below we describe how we find this portfolio
- First we build the so-called *active* portfolio, A , that invests in the n risky assets only (so not M)
- The weight of risky asset i in portfolio A is given by the formula:

$$w_i = \frac{\alpha_i / \sigma^2(\mathbf{e}_i)}{\alpha_1 / \sigma^2(\mathbf{e}_1) + \alpha_2 / \sigma^2(\mathbf{e}_2) + \dots + \alpha_n / \sigma^2(\mathbf{e}_n)}$$

- The weight of asset i increases in the asset's alpha, α_i , and decreases in the asset's residual standard deviation, $\sigma(\mathbf{e}_i)$

- Notice that when $\alpha_i < 0$, the weight of the asset is negative ($w_i < 0$), meaning that we *short* the asset
- Thus, in the active portfolio, A , we long assets that have positive alphas and short assets with negative alphas
- The index model applied to portfolio A takes the following form:

$$r_A(t) - r_F(t) = \alpha_A + \beta_A \times (r_M(t) - r_F(t)) + e_A(t)$$

where

$$\alpha_A = w_1 \times \alpha_1 + w_2 \times \alpha_2 + \dots + w_n \times \alpha_n$$

$$\beta_A = w_1 \times \beta_1 + w_2 \times \beta_2 + \dots + w_n \times \beta_n$$

$$e_A(t) = w_1 \times e_1(t) + w_2 \times e_2(t) + \dots + w_n \times e_n(t)$$

- In particular,

$$\sigma^2(e_A) = w_1^2 \times \sigma^2(e_1) + w_2^2 \times \sigma^2(e_2) + \dots + w_n^2 \times \sigma^2(e_n)$$

- The optimal risky portfolio, T , generated by the n risky assets and the market M can be viewed as a portfolio of the active portfolio A and the market M :

$$r_T(t) = w_A \times r_A(t) + (1 - w_A) \times r_M(t)$$

- If $\beta_A = 1$ (i.e., the active portfolio has the same beta as the market), the weight of the active portfolio is:

$$w_A^0 = \frac{\alpha_A / \sigma^2(e_A)}{(E(r_M) - e(r_F)) / \sigma_M^2}$$

- If the active portfolio has a beta of β_A then the weight of the active portfolio becomes:

$$w_A = \frac{w_A^0}{1 + (1 - \beta_A) \times w_A^0}$$

- We now calculate the Sharpe's ratio of the optimal risky portfolio T and compare it to the Sharpe's ratio for the market
- Risk premium of portfolio T is:

$$\begin{aligned} E(r_T) - E(r_F) &= w_A \times (E(r_A) - E(r_F)) + (1 - w_A) \times (E(r_M) - E(r_F)) \\ &= w_A \times \alpha_A + (1 + w_A \times (\beta_A - 1)) \times (E(r_M) - E(r_F)) \end{aligned}$$

- Similarly, the return volatility of portfolio T can be calculated with the following formula:

$$\sigma_T = \sqrt{(1 + w_A \times (\beta_A - 1))^2 \times \sigma_M^2 + (w_A)^2 \sigma^2(e_A)}$$

- The Sharpe's ratio of portfolio T becomes:

$$SR_T = \frac{E(r_T) - E(r_F)}{\sigma_T}$$

- If we plug the formula for w_A into the Sharpe's ratio for portfolio T , we obtain the following result:

$$SR_T^2 = SR_M^2 + \left[\frac{\alpha_A}{\sigma(e_A)} \right]^2$$

- The ratio $\alpha_A/\sigma(e_A)$ is called the information ratio of portfolio A
- The information ratio captures the excess risk premium from non-market risk per unit of non-market risk
- This equation says that:
 - The optimal risky portfolio T will always have a larger Sharpe's ratio compared to the market
 - The difference between the Sharpe's ratio of T and M depends on the active portfolio's information ratio
- Thus, the only reason why the optimal risky portfolio T is not the passive market index M is because of the alphas of the risky assets that make up the active portfolio, A

- Next we illustrate how to implement the Treynor-Black model with an example
- Suppose that we want to invest in the following financial assets:
 - Broad portfolio that captures the market: SPY (ETF tracking S&P 500)
 - Several stocks that we can monitor closely and that we believe are undervalued (positive alphas):
 - Manufacturing sector: Ford and GM
 - Communication sector: Verizon and AT&T
 - Retail sector: Walmart and Target
- Please download TreynorBlackdata.xls from Week 11 in Canvas